

#ASLI ENGINEERING

Hyperloglog Internals

SWIPE

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Hyperloglog Internals

Counting unique elements from a stream is a common usecase

A naive implementation

↳ insert all elements in a set

↳ count and return

1, 2, 3, 2, 3, 4, 1, 2, 3, 1

↓

{1, 2, 3, 4}

Using sets to do this is extremely expensive on the memory

So, can we approximate?

Flajolet-Martin Algorithm



- similar to Bloom Filter
- hash, byte array, quick computation
- storage required $O(\log m)$ where # unique elements = m

Core idea: uses position of rightmost set/unset bit to approximate

Number	Binary	ρ position of rightmost set bit
0	000	NA
1	001	0
2	010	1
3	011	0
4	100	2
5	101	0
6	110	1
7	111	0

$Pr[\rho = 0] = 4/8 = 1/2$

$Pr[\rho = 1] = 2/8 = 1/4$

$Pr[\rho = 2] = 1/8$

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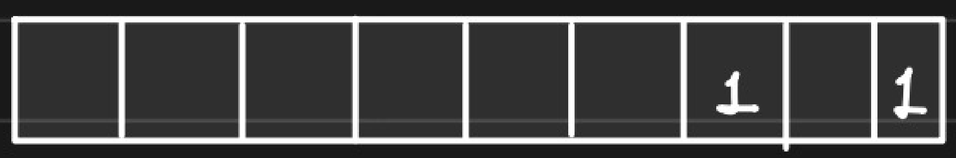
Thus we see that the probability of rightmost set bit drops by a factor of $1/2$ from LSB to MSB



For all incoming elements,

1. hash them out such that
   - it is an integer
   - distribution is uniform
2. record the rightmost set bit
3. cardinality = 2^b

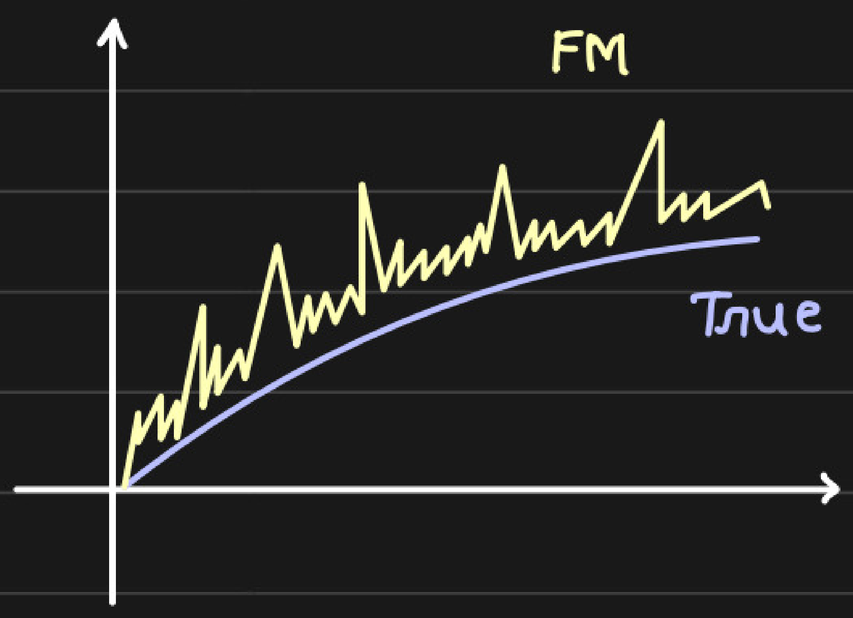
   where b is the rightmost unset bit



apple → 7 → 111 → 0

banana → 4 → 100 → 2

banana → 4 → 100 → 2



Why ?

if there were more than 'm' unique elements then the bit 'b' would have been set.

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